

HAT-P-20b

R-1194

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-20_250115 · created 2026-07-31T05:58:08+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460690.67638 +/- 0.00092 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1435 +/- 0.0046
Transit depth (Rp/Rs)^2	2.06 +/- 0.13 [%]
Orbital Inclination (inc)	88.88 +/- 0.48
Airmass coefficient 1 (a1)	1.1065 +/- 0.0095
Airmass coefficient 2 (a2)	-0.0911 +/- 0.0033
Scatter in the residuals of the lightcurve fit is	0.86 %
Best Comparison Star	None
Optimal Aperture	3.75
Optimal Annulus	11.62
Transit Duration (day)	0.0925 +/- 0.0017

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1435 ± 0.0046 vs 0.1284 (deviation 2.08σ)
Duration fit vs published	0.0925 d vs 0.0775 d (deviation 5.59σ)
Mid-time O–C	5.6 min
Depth SNR	19.8
Residual scatter	0.86 %

Light curve

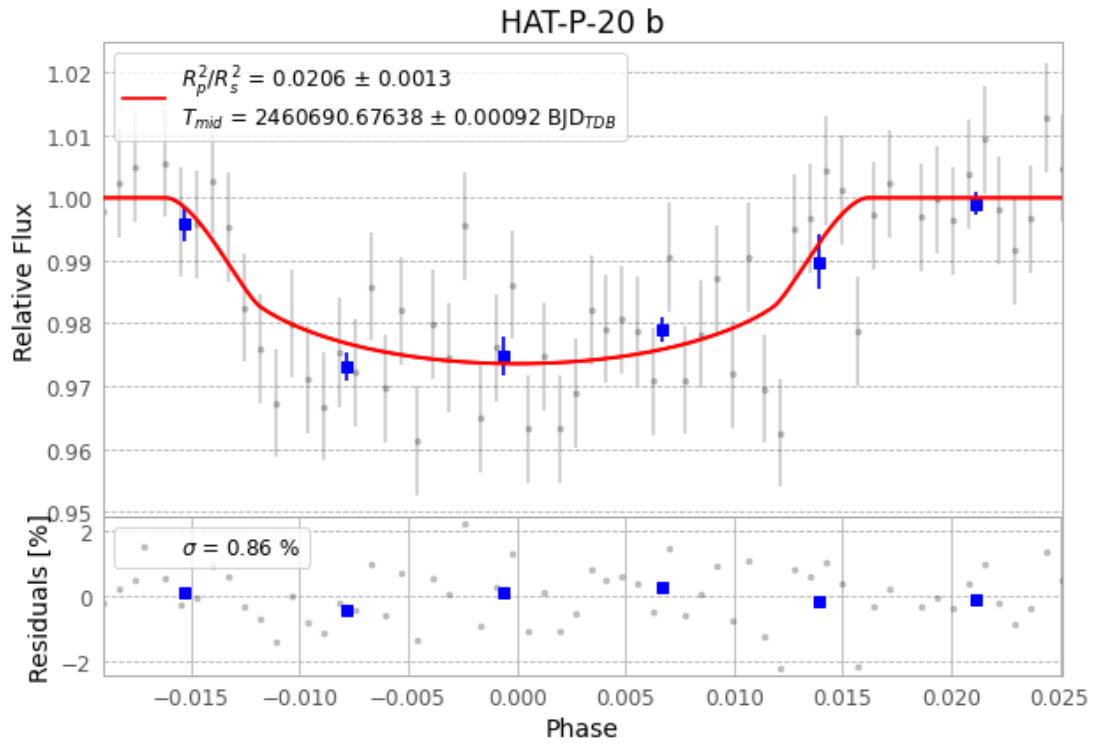


Figure 1 — FinalLightCurve_HAT-P-20 b_2025-01-14.png (SHA-256 f403bbd8147a8ab1...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [416, 187], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 cf1a880940b28f6a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

62 FITS frames (41 MB), HATP-20250115024516.FITS → HATP-20250115054815.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-20-250115.log (361b3165c71a...), FinalLightCurve_HAT-P-20 b_2025-01-14.png (f403bbd8147a...), FinalLightCurve_HAT-P-20 b_2025-01-14.csv (5e15498db538...)

Compute resources

Wall time 115.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 05:58Z.